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Nitte (Deemed to be University)

(Established under Sec. (3) of UGC ACT, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25EVT102– VEHICLE DYNAMICS

(For EVT)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

1. a) Explain vehicle fixed and earth fixed coordinator system.
 b) You are planning to buy a new mini-van to pull your boat trailer (see Figure.1) out to those long weekends at the lake. Although you like the new front-wheel-drive (FWD) vans available, you are not sure a FWD will be able to pull the boat up out of the water on some of the steep access ramps you must use.

Marks	BT*	CO*	PO*
8	L2	1	1,2

a) Derive the expressions for the maximum grade it can climb without wheel slippage (traction-limited gradeability) for this vehicle combination in a front-wheel-drive (FWD), rear-wheel-drive (RWD), and four-wheel-drive (4WD) power train.

(In the analysis it is reasonable to assume the longitudinal acceleration is zero, neglect rolling resistance, assume the boat is clear of the water so that there are no bouyancy forces on it, ignore any change in hitch height as the forces are applied, and use the small angle approximations.)

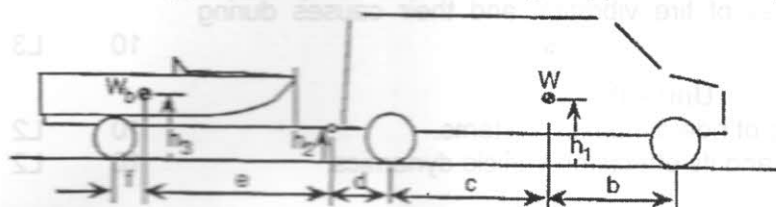


Figure. 1 Vehicle and Boat Trailer Combination

2. a) Derive the expressions for the dynamic axle loads acting on the front and rear axes of a vehicle during low-speed acceleration on a level road, and obtain the final expressions for W_f and W_r .
 b) Using derived expressions, calculate the maximum gradeability for the three combinations on a ramp with a coefficient of friction of 0.3, given the following information on the vehicles.

12	L4	1	1,2
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12	L4	1	1,2
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Van properties:

Front axle weight = 1520 lb

Rear axle weight = 1150 lb

CG height = 24.5 inches

Hitch height = 14 inches

Hitch rear overhang = 23 inches

Wheelbase = 120 inches

Combined boat/trailer properties:

Axle weight = 1200 lb

Hitch load = 250 lb

Wheelbase = 110 inches

CG height = 35 inches

8	L3	1	1,2
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Unit – II

3. a) Explain the layout and working of a vehicle powertrain with a neat sketch.

10	L2	2	1,2
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b)	Derive the expression for power-limited tractive force of a vehicle. Explain the significance of this limitation and discuss how it affects vehicle acceleration at different speeds.	10	L3	2	1,2
4. a)	Discuss the factors influencing braking performance in a vehicle. Using Newton's Second Law, derive the general equation for braking performance, explaining the relationships between vehicle weight, deceleration, brake forces, aerodynamic drag, and uphill grade. Furthermore, analyze how energy and power are involved during braking, considering the impact of wind resistance and brake system efficiency.	10	L3	2	1,2
b)	Derive the brake proportioning equation that describes the relationship between front and rear brake forces.	10	L3	2	1,2
Unit – III					
5. a)	A heavy truck weighing 72,500 lb rolls along 170 in Denver at a speed of 67 mph. The air temperature is 55°F and the barometric pressure is 26.01 in Hg. The truck is 8' wide by 13.5' high, and has an aerodynamic drag coefficient of 0.65. The truck has radial-ply tires. Calculate the aerodynamic drag, the rolling resistance and the road load horsepower at these conditions.	12	L3	3	2,3
b)	Explain the different types of vehicle resistance encountered during motion, and how each affects vehicle performance.	8	L3	3	2,3
6. a)	Explain the construction of radial and bias-ply tires and compare their performance characteristics.	10	L3	3	2,3
b)	Explain the different modes of tire vibration and their causes during vehicle operation.	10	L3	3	2,3
Unit – IV					
7. a)	Elaborate the fundamentals of ride dynamics systems.	10	L2	4	1,2
b)	Explain driveline excitation and its effects on vehicle dynamics.	10	L2	4	1,2
8. a)	Explain the low-speed cornering behavior of vehicles and derive expressions for the ideal steering angles using Ackermann geometry.	8	L2	4	1,2
b)	Derive the expression for the steer angle required at the front wheels during high-speed cornering, considering slip angles and cornering stiffness.	12	L3	4	1,2
Unit – V					
9. a)	Explain various types of suspension.	8	L2	5	1,2
b)	Explain the concept of Anti-Dive Suspension Geometry and its role in resisting braking-induced pitch.	12	L3	5	1,2
10. a)	Derive the expression for the pitch angle (θ_p) of a rear solid-drive axle vehicle during low-speed acceleration on a level road, explicitly considering the effects of anti-squat suspension geometry. Support your derivation with relevant equations and clearly explain the physical interpretation of each term.	10	L4	5	1,2
b)	Analyze and compare the pitch behavior of independent rear-drive, front-drive, and four-wheel-drive vehicles under acceleration. Derive the corresponding expressions for the pitch angle θ_p and discuss how suspension stiffness and drive torque distribution influence anti-squat and anti-pitch performance.	10	L4	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
